

★ WEEK 16 · PART 2 · PRACTICE

The Fair Comparison, Worked by Hand

A Three-Way Scoreboard, Walk-Forward, and a Leak You Can See

Grade three forecasts; test twice; catch look-ahead with numbers

Part 2's method, with the arithmetic shown. We grade three forecasts on held-out days, watch the simplest win, then see how one split can flatter - and how a single leaked statistic inflates a score it shouldn't.

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PART 2 · APPLICATIONS

Two Comparisons, Computed by Hand

Case A: grade naive, AR, and a mean forecast on five held-out days and let RMSE decide. Case B: see one split flatter a model, and watch a leaked statistic inflate a z-score. Then a Claude Code lab.

What Part 2 Told Us

Now we make it concrete - with numbers

1

Fit on the past, grade on the future

Reserve a test period, fit on the earlier data only, and score each model on the slice it never saw.

2

Walk forward so one split can't fool you

A single split can flatter by luck. Averaging over many out-of-sample forecasts is more honest - we'll see the numbers move.

3

Look-ahead bias fakes great results

Any statistic computed with test data leaks the future. We compute an honest score and a leaked one side by side.

Everything here is a calculator exercise: five held-out days, two test setups, one leaked statistic.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the scores by hand, step by step.
- Then read what the numbers really say.

What you should watch

- How the simplest forecast can win.
- How one split flatters a model.
- How a leaked statistic inflates a score.

A Three-Way Scoreboard

Grade three forecasts, let RMSE decide

We forecast a return three ways: the naive forecast, an AR(0.4) model, and simply predicting the long-run mean. Which is best on this asset? We grade all three on five held-out days and compute their RMSE by hand.

The three candidates

- Naive: repeat the last return.
- AR(0.4): scale the last return by 0.4.
- Mean: just predict the average (here, zero).

What we'll do by hand

- Forecast five held-out days each way.
- Square the errors and sum them.
- Turn each sum into an RMSE and rank.

Three Ways to Forecast

From doing nothing to a model

The two simple ones

- Naive: tomorrow equals today's return.
- Mean: tomorrow equals the long-run average.
- Neither needs any fitting on features.

The model

- AR(0.4): tomorrow = 0.4 times today.
- It uses one estimated coefficient.
- The 'sophisticated' entry in the race.

The Held-Out Test: Five Days

Forecast each day three ways

Day	$r(t-1)$	actual	naive	AR(.4)	mean
1	+0.6	-0.2	0.6	0.24	0
2	-0.4	+0.5	-0.4	-0.16	0
3	+1.0	-0.6	1.0	0.40	0
4	-0.8	+0.7	-0.8	-0.32	0
5	+0.5	-0.3	0.5	0.20	0

Goal: score each forecast by its squared errors, then RMSE. naive = $r(t-1)$; AR = $0.4 \times r(t-1)$; mean = 0.

Score All Three

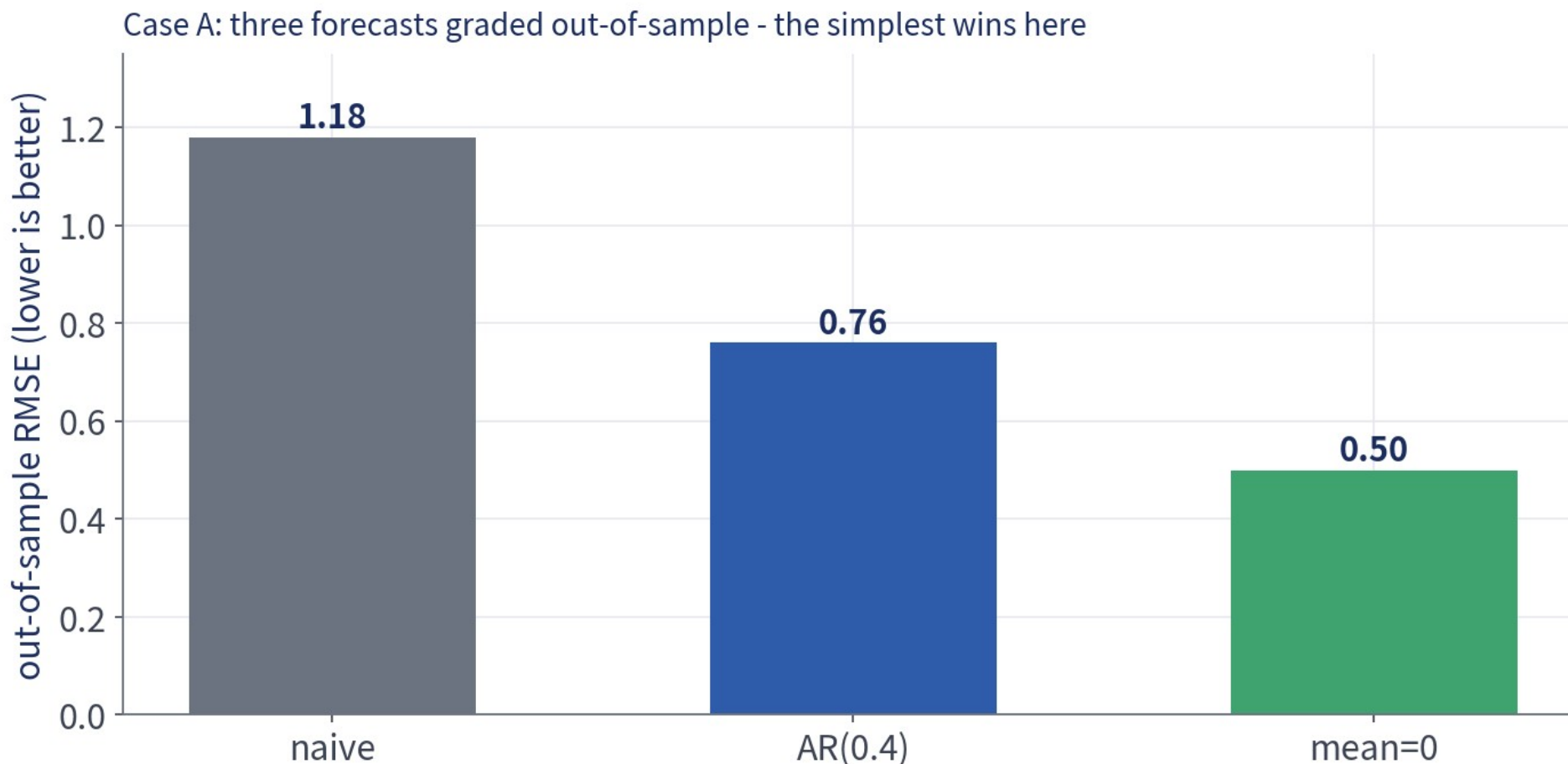
Squared-error sums to RMSE

naive: squared-error sum = 6.90 $\Rightarrow$ RMSE = $\sqrt{6.90/5} = 1.18$
AR(0.4) : squared-error sum = 2.92 $\Rightarrow$ RMSE = $\sqrt{2.92/5} = 0.76$
mean = 0 : squared-error sum = 1.23 $\Rightarrow$ RMSE = $\sqrt{1.23/5} = \mathbf{0.50}$
 $\Rightarrow$ the simplest forecast wins here - the opposite of Part 1

The mean forecast wins with RMSE 0.50 - the naive and the AR both do worse. Here, doing almost nothing is best.

The Three-Way Scoreboard

As a picture: the mean forecast has the lowest out-of-sample RMSE, the AR sits in the middle, and the naive is worst. On this asset the returns are close to unpredictable, so the plainest forecast wins.



Simple Wins Here

And that's the whole point

Compare to Part 1

- There, AR(1) beat the naive forecast.
- Here, the mean beats both.
- Different asset, different winner.

The lesson

- No forecast is best on every series.
- You cannot know the winner in advance.
- That is exactly why you always test.

Always Grade Against a Benchmark

Naive and mean are the bars

Why benchmarks

- The naive and mean forecasts cost nothing.
- For returns, they are often very hard to beat.
- They are the honest bar any model must clear.

The habit

- Report your model next to both benchmarks.
- If it can't beat them, prefer the simpler one.
- A model earns its keep only by scoring better.

One Split Can Lie - and Don't Peek

Two ways a test can mislead you

Two more traps, made numeric. First, a single train/test split can flatter a model just by luck; walk-forward averages that away. Second, a statistic computed with test data leaks the future and inflates the score. We compute both.

Trap 1: one lucky split

- A single split scores only a few points.
- It can land on an easy stretch.
- Walk-forward averages over many points.

Trap 2: a leaked statistic

- Standardizing with full-sample stats peeks.
- The 'test' data helped compute the scaler.
- The score is inflated, not earned.

Single Split vs Walk-Forward

Same model, two ways to grade it

Single split

- Fit once, forecast the last four days.
- Score those four out-of-sample errors.
- Fast - but only four points of evidence.

Walk-forward

- Forecast each day from the one before.
- Collect seven out-of-sample errors.
- More points, and a fairer average.

Grade the Same Model Twice

Single split vs walk-forward

single split (4 test points): $\text{RMSE} = 0.71$

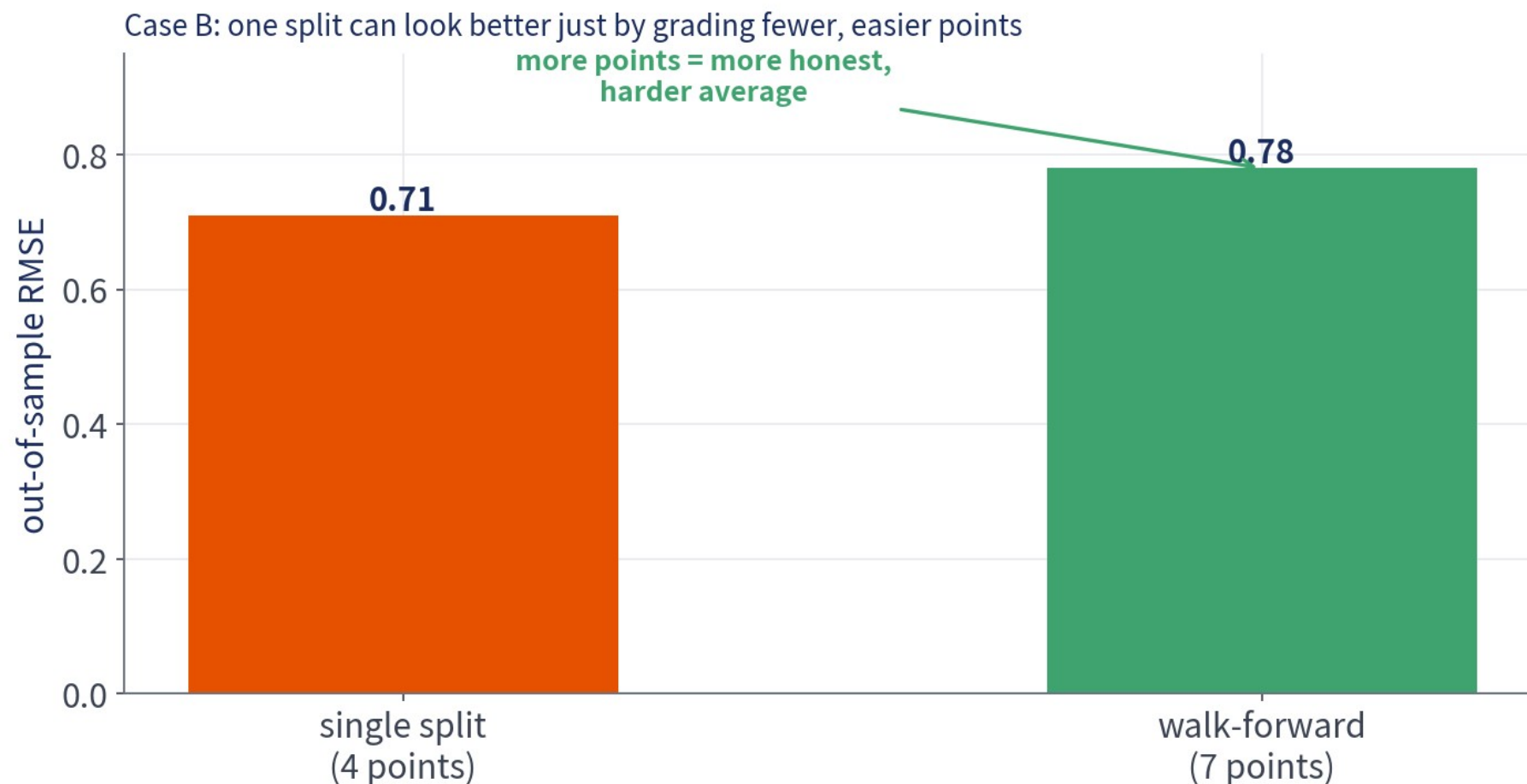
walk-forward (7 test points): $\text{RMSE} = 0.78$

⇒ one split looked better by grading fewer, easier points

The single split shows RMSE 0.71; walk-forward shows 0.78. The lower number wasn't skill - it just graded fewer, easier points.

One Split Flatters

The single split reports a lower RMSE than walk-forward for the very same model. It isn't better - it just graded four convenient points. Walk-forward's seven points give the more honest, and slightly higher, number.



Look-Ahead Bias, by the Numbers

One statistic can peek at the future

Look-ahead bias is subtle. Suppose you standardize returns into a z-score signal. If you use the standard deviation of the WHOLE series - including the test - your 'past' signal secretly knows the future. We compute the honest z and the leaked z and compare.

The honest way

- Compute the mean and sd on training only.
- Apply them to standardize the test points.
- Nothing from the future touches the scaler.

The leaked way

- Compute the sd over the whole series.
- That sd used the test data itself.
- Every z-score is quietly contaminated.

Honest z vs Leaked z

The same point, two standardizations

$$\begin{array}{ll} \text{train sd} = 0.66, & \text{full-sample sd} = 0.56 \text{ (secretly uses the test)} \\ \text{honest: } z = \frac{0.40 - 0.10}{0.66} = 0.46, & \text{leaked: } z = \frac{0.40 - 0.10}{0.56} = \mathbf{0.54} \end{array}$$

The leaked z (0.54) is larger than the honest one (0.46) because its denominator used the test data - a future it should never have seen.

What Actually Leaked

A small number, a big lesson

It looks harmless

- The two z-scores differ only a little.
- But the leaked one used the test to scale.
- Scaled across a whole backtest, it compounds.

Why it's dangerous

- Leaked signals look sharper than they are.
- The edge evaporates in live trading.
- Compute every statistic on training only.

The Common Pattern

Two comparisons, one discipline

Both cases, computed by hand, protect you from fooling yourself. Grade every forecast against simple benchmarks and let the numbers rank them. Average over many out-of-sample points, and compute every statistic on training data only.

The traps

- Assuming the fancy model wins.
- Trusting one lucky split.
- Leaking a statistic from the test.

The fixes

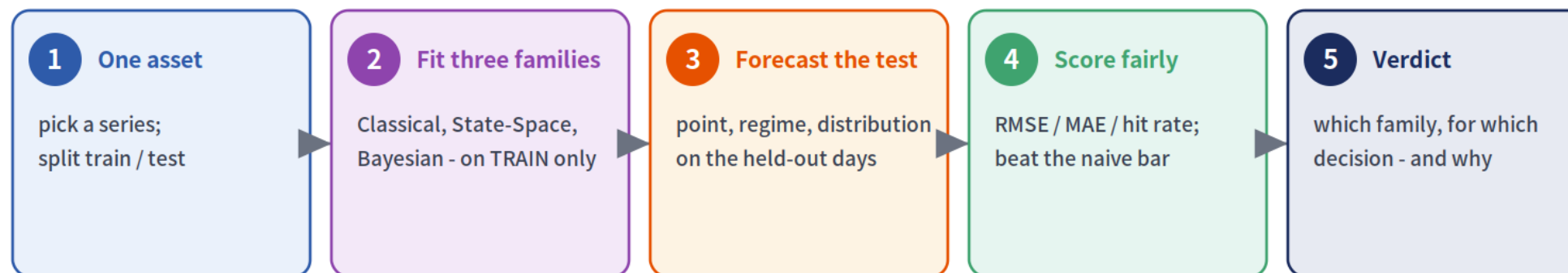
- Grade against naive and mean.
- Walk forward over many points.
- Fit every scaler on train only.

Where the Hand Calcs Fit

Case A is the 'score fairly' box; Case B guards the two honesty rules - many out-of-sample points, and no statistic from the test. Together they turn the capstone pipeline from a diagram into arithmetic you can check

The Capstone as a Pipeline

One asset in, three fitted families, one held-out grade, one honest verdict out.



The two rules that keep it honest

1. Fit on the past only - never let a model touch the test data before it is graded (no look-ahead bias).
2. Compare every family on the SAME held-out days, against the naive benchmark, with the same metrics.

The verdict is decision-driven: the winner is the family whose output your decision actually needs.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both comparisons in code, then extend them to a real asset.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - scoreboard:

- * take the 5-day table; form naive, AR(0.4), and mean forecasts; print squared-error sums;
- * compute and rank the three RMSEs.

CASE B - honesty:

- * grade one AR model by a single split and by walk-forward; print both RMSEs;
- * standardize a test point with train-only sd and with full-sample sd; print both z-scores.

Match every printed number to the slides, then repeat on a real return series.

Check against the slides

- RMSE naive 1.18, AR 0.76, mean 0.50.
- Single split 0.71, walk-forward 0.78.
- Honest z 0.46, leaked z 0.54.
- Simple wins; one split flatters; leaks inflate.

Deliverable: `part2_practice.py` whose printed numbers match the hand calculations, then a real asset.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why did the mean forecast win Case A?

Lowest out-of-sample RMSE on this asset

Q2 Why was the single-split RMSE lower?

It graded fewer, easier points

Q3 Why was the leaked z-score larger?

Its sd used the test data

All three come from the scoreboard, the walk-forward comparison, and the leaked standardization.

Three Pitfalls to Avoid

Where comparisons mislead

Common mistakes

- "The model must beat the naive forecast."
- "One good split proves the model."
- "Standardizing on all the data is fine."

The reality

- Sometimes the mean wins - so always test.
- One split can flatter; walk forward.
- Full-sample stats leak the future - use train only.

Two Comparisons to Remember

Part 2 applications in one screen

1

Grade three forecasts; let the number decide

Naive, AR(0.4), and a mean forecast scored RMSE 1.18, 0.76, and 0.50. The simplest won on this asset - the reverse of Part 1 - which is exactly why you never assume and always test against a benchmark.

2

One split can flatter; walk forward

The same model scored 0.71 on a single split but 0.78 walk-forward. The lower number wasn't skill; it just graded fewer, easier points. Many out-of-sample points give the honest average.

3

A single leaked statistic inflates the score

Standardizing with a full-sample sd (0.56) instead of the train sd (0.66) turned an honest z of 0.46 into a leaked 0.54. Compute every statistic on training data only.

Next — Part 3: the final project end to end, from comparison to a trading decision, and the course close.